

Utilising eye-tracking data in retailing field research: A practical guide

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Abstract

Eye-tracking offers many insights into shopping cognition by capturing proxies to attention patterns, visual processing, and decision sequences. While lab experiments using simulated shopping tasks are becoming more common in retailing research, field research using eye-tracking technology remains scarce despite higher ecological validity. This paper advocates expanding eye-tracking adoption into brick-and-mortar contexts and provides guidelines for researchers interested in conducting such retailing research. The authors discuss different data types that can be used to understand in-store shopping behaviours and argue why eye-tracking offers a compelling opportunity to decode consumer decisions in real-world complexity. Next, they review current literature, highlighting good examples of existing field studies and areas of opportunities for new research. Finally, the authors provide stepwise best-practice guidelines for executing impactful, triangulated field research across the retail landscape using eye-tracking. The paper presents a call to action for expanding eye-tracking research beyond the lab and into real-world retail aisles.

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Introduction

In recent years, the study of shopper marketing has gained new interest. This renewed interest can be attributed to several reasons, such as shoppers' unexpected return to physical stores after the pandemic (Weinswig 2023), the increased digitalisation of physical stores (e.g., Grewal et al. 2023a), and the change in shopper behaviour due to various digital devices out of the retailer's control (e.g., Grewal et al. 2018). Retail practitioners often talk about the importance of experiential marketing. A related concept that take a more theoretical approach to the same phenomenon is sensory marketing. Independent of the practical or theoretical point of departure, the way customers act and react to various sensory stimuli

in retail environments often needs to be measured in detail, and this can be complicated with all the noise that shoppers incidentally encounter whilst shopping in an actual store.

Traditional methods for tracking in-store shopper behaviour include surveys and observational studies. In early shopper marketing studies, researchers discretely tracked shoppers by taking notes of their path through the store, the stimuli they looked at, stopped by, or picked up, and other observable facts such as their companions, age, and gender (e.g., Granbois 1968). In some cases, the research question concerned something other than understanding what a shopper spontaneously paid attention to while browsing an entire store. Instead, the research question involved a more local behaviour, such as what a shopper fixated on within a particular category or shelf section. Early studies of this more delimited character were also done manually (e.g., Hoyer 1984). Whilst early shopper research has greatly benefitted the field of retailing, traditional methods have limitations in capturing nuanced, real-time behavioural data. Surveys rely on shoppers' self-reported perceptions and memories, which can be biased or incomplete — and observational studies using manual note-taking struggle to capture fine details about shoppers' attention and decision processes.

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Technological advancements have changed the observation methods for studying shoppers' movement patterns through the stores. Some of the technologies used are CCTVs (Gil et al. 2009), RFID chips placed on shopping carts (Hui, Bradlow and Fader 2009), and small cameras put on headsets (Sorensen 2008). These types of observations provide richer data for retailing researchers as they try to understand shopper behaviours. Yet, they lack a vital element of the decision process – precisely measuring what catches shoppers' visual attention. Visual attention is significant for a wide range of research questions in retailing research, as visual attention is one of the essential prerequisites for most shopping decisions. Understanding precisely and objectively what shoppers look at is critical for retailers to optimise marketing strategies and store layouts.

As such, in the past couple of decades, eye-tracking research has become the dominant method for this kind of observation, as it overcomes many limitations of traditional tracking methods (e.g., Grewal et al. 2020). Eye-tracking technology precisely tracks where shoppers look and approximates where they focus their attention. When shoppers move around freely in retailing contexts, data on what draws their attention can easily be recorded using ambulatory eye-tracking glasses (Grewal et al. 2018). When shoppers are standing still or sitting, in contrast to freely moving around in an open space, a stationary eye-tracker is an alternative that is usually easier to code. These stationary eye-trackers are also particularly useful in studying online shopping, as such data can be collected and analysed relatively simply compared to ambulatory eye-tracking.

This paper explores how eye-tracking addresses gaps in traditional shopper tracking, providing researchers and retailers with a more accurate, granular understanding of in-store behaviour. Examples illustrate how fixing these blind spots with eye-tracking can enhance store layouts, assortments, shopper experiences, and online retailing. We advocate for broadly expanding eye-tracking methods in field-based retailing research to match the proliferation of laboratory studies that have become more common over the past 25 years. First, we elaborate on foundational motivations explaining why eye-tracking data offers an unparalleled window into shopping cognition. Next, we survey current publication trends and make the case that retail field investigations remain comparatively scarce despite their unmatched authenticity. We then provide best-practice guidelines for implementing impactful eye-tracking projects across diverse brick-and-mortar contexts. Finally, we issue targeted recommendations for high-potential research avenues deserving of dedicated attention from consumer behaviour scholars. We aim to spur the next quarter century of eye-tracking research beyond the lab and into the aisles.

Methods for studying shopper behaviours

Field studies and ecological validity. To fully understand shopper decision-making, conducting studies in the natural environment where the shopper makes the decisions, (i.e., in the stores,) is crucial. This is so for at least two reasons. First,

a store is not a controlled environment. Multiple external stimuli will influence shopper's behaviour, such as the fact that the shopper only visits a limited part of the store (Sorensen 2008), the competition for attention from, for instance, other products (Phillips et al. 2015), signage (Roggeveen et al. 2016), the store atmospherics (Baker et al. 2002) and other shoppers (Hui and Bateson 1991). Moreover, the shoppers' goals (Wedel and Pieters 2008), knowledge of the store (Park, Iyer and Smith 1989), and stress level (Ahlbom et al. 2023) also could have a more significant influence in a natural setting than in a controlled lab setting. Second, many laboratory studies are based on forced exposure. Hence, in a lab study, there is limited competition for attention so that the shopper can go straight to the evaluation phase involving analytical processing. However, while browsing a store, a shopper undertakes a broader range of and more holistic visual processing. During the browsing phase, as much as 99.7% of the assortment could be disregarded (unpublished material based on more than 1000 eye-tracking videos) before any attentive processing, such as an evaluation, takes place (Janiszewski, Kuo and Tavassoli 2013). Given this, but depending on the research question, field studies on shopper decision-making give a more ecologically valid picture.

Data sources to understand shopping behaviours. Researchers have examined in-store behaviours in various ways, all with various strengths and weaknesses (see Table 1). Early studies of shoppers in stores often relied on manual data collection in the form of unobtrusive shopper observations. The observations were made by research assistants dressed up as sales assistants working for the retailer (Granbois 1968), and the data was collected with the help of paper and pen. Manual observations lack detail because it is tricky to know exactly what a shopper is looking at (e.g. if the shopper is looking at a price tag or a package) and how much time a shopper is spending paying attention to various stimuli.

To cope with these problems, new data collection methods have been gradually developed and implemented in the field. RFID tags have, for instance, been put on shopping carts and shopping baskets to measure shopper flows throughout the stores (Hui, Bradlow and Fader 2009). These trackers are advantageous for collecting large amounts of data, but they, too, suffer from challenges with precision. This is because it is common for shoppers to leave their cart or basket on the shop floor while they walk into an aisle to pick up a product (Larson, Bradlow, and Fader 2005; Sorensen 2008). Another method is to use video cameras to capture shoppers' movements (Gil et al. 2009). While this method can be sufficient if the purpose is to study shopper movements to generate heatmaps in one department or for one shelf, it can be more problematic if the researcher wants to study areas of a store that would mean that the researcher would have to synchronise shoppers being captured by different cameras (Newman 2002). While the method may be enough to reveal a shopper's path and where a shopper stops, it is insufficient to capture precisely where the shopper is looking on a shelf. Yet another method is the so-called headcam, a small camera that can be placed behind the ear of the shopper,

Table 1
Comparison of some data collection methods for tracking in-store shopping behaviours: strengths and weaknesses.

Method	Strengths	Weaknesses
Surveys	• Low cost • Easy to administer • Flexible analysis	• Rely on memory/perceptions • Prone to bias
Manually tracked observations	• Capture actual behaviours • Context rich	• Labour intensive • Limited detail
RFID Tracking	• Large data sets • Automated tracking	• Imprecise (when shoppers leave carts) • Limited to cart movement behaviours
Video/CCTV monitoring	• Behaviour tracking • Spatial patterns	• Complex video synchronisation • Privacy concerns
Eye-tracking –Screen-based lab studies	• Controlled setting • Easier analysis compared to ambulatory eye-tracking	• Forced exposure • Less realism
Eye-tracking –VR-based lab studies	• Controlled setting • Easier analysis compared to ambulatory eye-tracking • Better realism for bricks-and-mortar simulations than screen-based	• Resource-intensive VR simulations • Lacks input from senses other than vision • Artificial
Eye-tracking – Field studies	• Precise attention data • Authentic behaviours • Context rich	• Costly equipment • Time-consuming data collection and analysis • Sample size limitations

much like a headset (Sorensen 2008). While headcams sort out many of the issues with the earlier methods, they only capture head movements and the direction of the shoppers' gaze. This leaves us with the best method for capturing shoppers' attention so far – the eye tracker. Eye trackers do not only capture the gaze but also the fixations and the saccades. Fixations refer to when the pupil stands still, and saccades refer to the movements between fixations. Metrics such as the first fixation, number of fixations, length in time of the fixations, length of the saccades (i.e., the distance between fixations) and scan paths (i.e., the sequence of the fixations) can only be captured with eye-tracking.

In bricks-and-mortar retailing research, ambulatory eye-trackers ("eye-tracking glasses") are often a particularly suitable choice. Ambulatory eye trackers are glasses with built-in cameras that record all shopper's fixations while wearing the glasses. These have the benefit of collecting data about the shoppers' eye movements without interfering with the shoppers' natural behaviours. Ambulatory eye trackers capture a shopper's path through the store, direction of gaze, focus of attention and time spent on all these behaviours. For field studies in e-commerce settings, stationary eye trackers may provide a more resource-efficient alternative. These eye trackers are mounted on a screen or utilise web camera recordings (which are cheaper but come at the cost of lower accuracy) so that participants' eye movements match what is shown on the screen. This type of data can be more easily coded by software than ambulatory eye tracking can. A third alternative is virtual reality (VR) eye tracking, which records what participants see in a simulated store setup (Meißner et al. 2019). Whilst not without challenges, as we discuss later, there are several strengths in collecting fine-grained, precise measures of where and how people look due to how shoppers make their in-store decisions.

Visual perception and shopper decision-making

Data about shopper movement in general and eye movements in particular are essential keys to the study of shopper decision-making, but it is usually insufficient. It is also essential to understand shoppers' upstream processes to make meaningful interpretations of the eye movements. The study of not only visual attention but also visual perception is critical because most in-store decisions are mixed in the sense that they build on a combination of (1) stimulus input and (2) memory retrieval that, in turn, might have been activated by an in-store stimulus (Lynch and Srull 1982). In support of this idea, studies have found that shoppers combine memories from previous shopping trips with visual stimuli from a current trip to effortlessly detect their searched-for product among hundreds of alternatives within a minimal time frame (Dickson and Sawyer 1990; Hoyer 1984). Hence, a very short fixation or a limited number of fixations could be signs of perceptual fluency (brand loyalty) or a lack of interest (Shapiro 1999).

In a sense, it could be argued that the study of attention is "hollow" without considering the context. For instance, it has been found that the number of facings of a product substantially impacts the product's evaluation, and this effect is fully mediated by visual attention (Chandon et al. 2009). However, the same study also found that the very same eye movements need to be given different interpretations depending on both memory factors (e.g. familiarity with the brand) and external stimulus factors (e.g. the shelf height and the product's horizontal position). That means that even though attention – counted as the number of fixations on a product – was found to be necessary for a purchase, eye-tracking measures needed to be combined with other data to be assigned the correct interpretation.

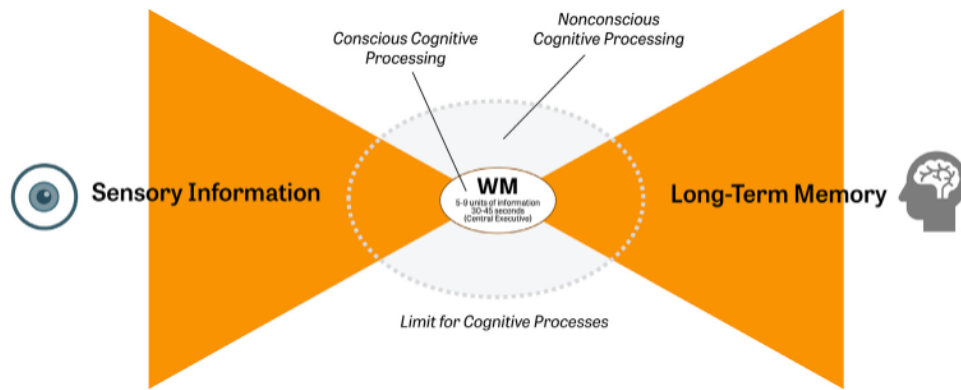


Fig. 1. A Model for in-store decision-making based on a limited capacity Working Memory (WM) and a Bottom-Up (Visual) focus.

Table 2
Capacity of Different Senses.

Sensory nervous system	Capacity, bits/second	% of all information the brain receives
Sight	$2 \times 3000,000$	83%
Hearing	$2 \times 20,000$	11%
Touch	10	1.5%
Taste	10–100	3.5%
Smell	2000–200,000	1%

Source: Nordfält (2011).

A model built on sensory input mixed with memorised input is a good starting point for interpreting eye-tracking data. Theoretically, this combines the processing of external stimuli and stored memories in the working memory (Baddeley 2021). Parallel preconscious streams of perceptual and conceptual data are combined into a whole in the central executive of the working memory. This is depicted in Fig. 1. To understand the processing of sensory information (the left side of Fig. 1), eye-tracking provides an excellent opportunity, especially regarding such visual processing that is nonconscious. Other data sources, such as questionnaires and interviews, mainly focus on conscious processes and directly auxiliary feelings and cognitions (as depicted by the limits for conscious cognitive processing). They are stored in the long-term memory and are recollected during data collection, such as when shoppers are intercepted at the end of a half-hour shopping trip and asked about thoughts such as feelings and experiences regarding the shopping trip.

The promise of eye-tracking data relates to the undisputed fact that visual perception has the most significant influence on in-store decision-making of all the human senses. A comparison of the bandwidth of the various senses reveals that the human eye is by far the most powerful human sense. Depending on the source, as much as 83% of all sensory input has been attributed to vision (Table 2). This comparison may not be fair since the different senses do not only differ in their bandwidth but also in other qualities. For instance, senses like sound and smell are captured in 360°, meaning that independent of the shoppers' locations and directions, they will be exposed to sounds and smells as opposed to visual stimuli

that need to be within their fields of vision. Nevertheless, the study of shopper behaviour is highly tied to the study of visual attention.

Based on the above, we argue that eye-tracking, in combination with methods to capture upstream cognitions, is the best way to study shopper behaviour to date.

Applications of eye-tracking in retailing field research

While eye-tracking offers several opportunities for retailing researchers, it remains a relatively young marketing and retailing research method. That said, a growing number of researchers have adopted eye-tracking data to answer research questions to various degrees, highlighting several opportunities.

Lab studies. Most retailing research and adjacent research areas utilise eye-tracking in controlled, computer-based environments, often published in the more general marketing literature. This is suitable for many research questions that focus on understanding particular marketing elements and how they can be broadly optimised for retailers and marketers. For example, much eye-tracking research examines marketing elements like package design elements such as labels (e.g., Werle et al. 2022, Sielicka-Różyńska et al. 2021), advertising (e.g., Venkatraman et al. 2021, Yin et al. 2021), and website and search optimisation (e.g., Shi and Trusov 2021, van der Lans, Pieters, and Wedel 2021, Biswas et al. 2023). These studies are often experimental, and participants see one of several versions of a product packaging, website, or similar. Examining more basic and tightly controlled reactions to various marketing stimuli is easy through computer-based eye-tracking, which offers practical advantages in administering experiments. Researchers can readily manipulate variables, quickly collect data, automate most data coding, and achieve a high degree of control over the testing environment. However, this comes at the cost of realism, ecological validity, and generalisability to real shopping contexts. Visual shopping processes are often divided into three stages (Russo and Leclerc 1994). The three stages are browsing (a holistic process where many sources of data are processed effortlessly in parallel), evaluation, and verification (analytic,

effortful, and conscious processes). In lab studies, the browsing stage is, at best, partial. Achieving natural exposure to shopping is tricky for retailing researchers, as participants' eye movements would have to be tracked over an extended period of time to allow them to naturally decide to visit e-commerce websites rather than being instructed to do so. While some marketing research firms have experimented with tracking web browser users' eye movements over long time periods, this approach, which would require individual eye trackers or specially constructed computers, is likely unrealistic for most researchers at present.

Some researchers have suggested utilising virtual reality alternatives to the more traditional screen-based eye-tracking studies (Meißner et al. 2019). This eye-tracking type is a midpoint between a regular lab study and a field study, as it allows participants to enter a virtual store. This increases ecological validity substantially while making it easier to code for researchers. However, it comes at the cost of somewhat complicated setups of the virtual environment to represent a real-world setting and lack of natural movements and tactile inputs. This alternative may be suitable in specific situations when the researchers want to have tight control over experimental conditions without losing too much ecological validity.

Field studies. Despite more logistical challenges, in-store studies that utilise ambulatory eye-tracking data offer advantages in achieving ecological validity, which is often lacking in lab-based studies (Otterbring et al. 2023). The studies that have utilised ambulatory eye-tracking data in retailing tend to utilise in-store, brick-and-mortar environments. This enables the examination of complex in-store consumer behaviours and unplanned exposures absent in lab studies. For example, field experiments can capture shoppers' visual attention patterns in response to aisle layouts (Chen et al. 2024), displays (Londoño and Ruiz de Maya 2022), visual biases (Chen et al. 2021; Ladeira, de Oliveira Santini, and Pinto 2022), different types of product assortments (Ladeira et al. 2023) signage and other environmental factors (Huddleston et al. 2023).

Field studies utilising eye-tracking data can be helpful for both exploratory, descriptive and causal research designs. One example of a descriptive research study is that of Chen et al. (2021), where the researchers examined lateral and vertical biases in shoppers' attention. They intercepted shoppers before they entered a supermarket and asked them to "shop as normal" but with ambulatory eye-tracking glasses. After coding up the data, the authors explained where shoppers pay more or less attention on the shelf and aisle, a bias that has been discussed for years but has never been examined with such fine and detailed data about shopper attention. In another example, Grewal et al. (2023b) used a large set of eye-tracking videos in an exploratory fashion to understand how different types of in-store communication on in-store screens correlate with what shoppers buy. In this study, the authors correlated what content shoppers saw on the screen with the receipt they provided at the end of the shopping trip without experimental manipulation. This type of descriptive study can be a valuable complement to better understanding

the experimental findings from other studies with causal research design.

Other studies, such as Grewal et al. (2018), include experiments where shoppers are given different instructions before entering the store. In that article, shoppers were randomly encouraged to use or not use their mobile phones while shopping, and the comparison between these two groups showed that shoppers who used their phones looked at more products and spent more money. This finding was complemented with a post-shopping survey where the process mechanism of distraction was measured as another source of data to understand the upstream processing that was observed behaviourally. It highlights the benefits of triangulating data to understand better how shopping behaviours are formed.

Despite the promise of ambulatory eye-tracking data, far fewer retailing researchers have published research using it. This may be due to resource constraints. The equipment is many times more expensive to buy, and data collection is also slower, resulting in resource implications for the researcher's time or funds to pay research assistants. It may also have to do with the need for more coding. Unless the researcher is interested in a part of a store (e.g., a specific endcap) that can be automatically coded, coding ambulatory eye-tracking still requires significant manual labour. Another reason could be that many researchers need access to adequate research venues, such as real-life stores or behavioural labs with the facilities to recreate quasi-real retail environments.

A recent positive trend is the increasing adoption of ambulant, field-based eye-tracking methods to study retailing environments, albeit from modest levels. Unlike in the general marketing literature, which predominantly uses lab studies, most newer retailing-specific research that focuses on retailing-specific topics includes at least one in-situ experiment in a store setting, highlighting how essential field studies are to understanding the field of retailing. Despite this, the number of retail research utilising eye-tracking in many top-tier journals remains relatively low. This suggests that the mere use of eye-tracking to contribute to a paper may be insufficient for top-tier publications. Again, while intriguing, the data must be embedded in a broader research narrative advancing understudied areas or explaining understudied areas. Eye-tracking should not be used because it is novel or "cool" but strategically positioned within a mix of methods. Moving forward, we encourage retailing research projects that harness eye-tracking to tackle open questions, not just low-hanging fruit that have become apparent due to access to an eye-tracking kit. This will strengthen the credibility and visibility of eye-tracking in premier marketing and retailing journals. The goal should be impactful scholarship, with eye-tracking as an instrumental ingredient, not the sole focus.

Strengths with eye-tracking for retailing research

Field-based eye-tracking technology offers significant and unique strengths for advancing retail research across industry and academia. Gaining visual attention data directly from

shoppers in real-world environments provides extensive and deep insight into the consumer decision journey.

First, eye-tracking offers valuable behavioural data for explaining reactions, including difficult-to-measure nonconscious responses. Shopping is fundamentally a visual process - without seeing, we cannot evaluate products for purchase. As the maxim states: “you buy what you see.” Eye-tracking traces this visual attention and comparison process in situ, illustrating the subtle factors that shape purchase choices. With its fine-grained gaze data, eye-tracking can proxy nonconscious processing unobtainable through other means. This can be elements such as heuristics activated by specific cues in the store environment or sequential processing due to priming and mere exposure effects. Thus, it adds a powerful physiological dimension to analysing shopping journeys, potentially bridging many gaps in current understanding.

Additionally, the versatility of eye-tracking enables both qualitative discovery and quantitative modelling within a single method. Although this paper has focused primarily on quantitative methods and analyses, combining eye-tracking data with think-aloud protocols or other qualitative approaches allows rich journey mapping. As an unobtrusive data collection technology, it captures highly detailed shopper behaviours with minimal interference or bias.

Finally, eye-tracking intersects strongly with industry priorities and academic trends. Retailers desire greater clarity on why certain in-store elements succeed or fail in lifting satisfaction and loyalty. The visually engaging data gives clear strategic direction while satisfying academic curiosity about the psychology underlying retail contexts. As reviewers and partners convey, eye-tracking demonstrates ambition and commitment to furthering progressive research. As adoption continues, ambulatory eye-tracking is poised to transform retail research - uncovering new insights into the drivers of shopping behaviour through uniquely potent data. The strengths show immense promise for understanding and influencing dynamic consumer choice.

Weaknesses with eye-tracking for retailing research

While eye-tracking technology offers valuable insights into shopper attention and behaviour, we must consider its limitations when applying it to retailing research. First and foremost, we should examine whether it helps answer the specific research question or provides only ancillary details that are “nice to know” rather than substantively advancing our understanding. For example, when exploring the drivers of satisfaction with a service interaction and intentions to return to the store, granular eye-tracking data from the interaction itself may not add significant explanatory power over simply surveying shoppers afterwards about their perceptions. For research questions that go beyond mere attention, behavioural data from eye-tracking often needs to be combined with questionnaire data. That way, the researcher understands why shoppers pay attention to particular aspects of the stores and how this influences their internal state and decision-making.

Second, we must remember that eye movements and gazes are valuable proxies for attention allocation and other cognitive processes, but they do not directly measure mental processes. Shoppers may passively gaze at an area without deeply processing or internalising the information. Several algorithms have been developed to parse eye-tracking data, setting thresholds for distinguishing intentional attention versus random micro-gazes, but these classifications remain approximations. Relatedly, a shopper may extensively scan a visually complex assortment yet still walk away with limited memory and perception of the available variety because excessive information overwhelms our mental filters. For example, studies have found that perceptions of variety differed little between extensive and sparse assortments when eye-tracking patterns were equally extensive (Kahn and Wansink 2004). The indications that complex assortments reduce fluency results were later validated using eye-tracking (Orth and Crouch 2014).

Third, we must consider the potential for observer effects and biased behaviours when shoppers know they are being recorded. Most grow comfortable wearing eye-tracking glasses after sufficient time, as confirmed by those who forgetfully walk out of the store still wearing the expensive equipment! Nevertheless, any observation can alter behaviour - shoppers may spend more effort appearing attentive to promoted products if aware of the recording. Comparing recordings to incognito remote behavioural observations can help quantify and mitigate potential biases.

Finally, research-grade ambulatory eye-tracking equipment remains highly costly to obtain, operate, and analyse. All in-store field studies demand significant resource investments over surveys or lab studies. Adding intensive video data generation amplifies costs multiplicatively, especially concerning data coding. This often has to be done by assessing the videos frame by frame. However, we believe that more automated coding of ambulant eye-tracking videos, possibly aided by AI, will become accessible in the not-so-distant future. Pursuing eye-tracking thus requires carefully assessing desired benefits against sizable resource trade-offs. Eye-tracking desires may outweigh realistic capabilities for some research aims or target sample sizes. Other methods like shop along or trace measurements may more pragmatically serve specific goals. Lab-based eye-tracking data may also complement other in-store data collection methods to get at least some indication of the underlying attentional processes. However, this approach significantly reduces the ecological validity as it is so tightly controlled and often reminds participants that they are being observed.

When thoughtfully and transparently applied, eye-tracking offers remarkable visibility into shoppers’ attention patterns, choice processes, and behavioural responses. However, it can only solve some questions in retail research. By acknowledging its meaningful but definable limits, we underscore analysis integrity while retaining due optimism about eye-tracking’s substantial and growing value for advancing shopper understanding. Researchers aiming to deploy this powerful capability must first critically examine if it can genuinely answer the

research question asked over more straightforward methods while proactively quantifying and mitigating observer biases. If these criteria are met, eye-tracking delivers multidimensional insights unavailable through other means.

Opportunities for future retailing field research with eye-tracking

Expanding beyond grocery contexts. While most prior field research utilising eye-tracking has focused on grocery store environments, examining shoppers' visual attention and behaviour across other retail categories would be valuable. Speciality retailers like fashion, electronics and home goods likely elicit different browsing strategies and attention patterns compared to packaged goods. For instance, how do shoppers visually evaluate subjective aesthetic attributes when assessing fashion items? How might they compare products on complex features like electronics? Researchers could explore differences across retail segments to reveal new contextual insights.

Understanding general in-store biases. Eye tracking could map movement patterns and visual biases independent of specific categories. Identifying “hot” and “cold” spots that influence traffic flows and shopper navigation tendencies has significant implications for store layout optimisations, shelf space planning and slotting decisions. Cultural differences in baseline attention patterns are another area that warrants dedicated examination, especially considering that most eye-tracking field research is concentrated in a handful of countries. Could it, for example, be that what is “hot” and “cold” spot and other visual biases differ between a market where shoppers read from left to right (e.g., English-speaking countries) or right to left (e.g., Arabic-speaking countries)?

Tracking shopper navigation and decisions. Retailers must understand how shoppers navigate stores visually to inform layouts, displays and signage strategies. Do shoppers use gondola ends or signpost brands for wayfinding? Can eye movement sequences reveal decision stages and cross-category comparison priorities? Tracing visual transitions between the first, second and *n*th products viewed can optimise adjacencies to boost cross-selling.

Bridging physical and digital channels. Despite e-commerce's prominence, field studies tracing online visual behaviour remain scarce. For example, work suggests substantial differences from physical stores in how shoppers perceive the assortment (Kahn 2017). However, this could be explored in more detail. Multi-method research examining entire digital shopping journeys is needed to pinpoint factors influencing search processes, site interactions and purchase conversion. Cross-effects between physical and digital channels also deserve attention as omnichannel retail expands.

Utilising qualitative methods in eye-tracking research. Finally, a significant yet underutilised approach in eye-tracking research involves qualitatively analysing the rich visual footage through methods like ethnography and thematic coding, possibly also using the sound recording most of the ambulatory eye-tracking devices record. While current appli-

cations focus heavily on quantifying eye gaze metrics, play-by-play videos tracking complete shopping journeys enable valuable qualitative insights into consumer psychology and decision motivations, that are challenging to capture through numerical data alone. For example, researchers could inductively code videos to identify key themes, patterns, and variability in attention across shoppers when comparing products. This complements statistical findings on gaze duration and allows connecting fixations to underlying meaning. Despite promising fits, few studies leverage these qualitative capacities of eye tracker footage - offering low-hanging fruit for an impactful mixed-method study.

In conclusion, the field-based eye-tracking methodology has immense untapped potential for advancing retail consumer behaviour research. As equipment costs decrease and analytical techniques improve, we hope more scholars undertake field studies to address these open research questions. Expanding our empirical understanding of in-store decision-making will have important theoretical and practical implications. The following section provides a hands-on guide to how an eye-tracking study could be conducted in a field setting.

A hands-on guide to utilising eye-tracking in retailing field studies

Eye-tracking methods have the potential to offer valuable insights into customer attention and engagement when incorporated into field research in retail settings. We now suggest a stepwise approach to conducting field research to answer targeted research questions. We will cover determining when eye-tracking is most valuable, planning an appropriate sample size, practical considerations for study setup and execution, analysing the resulting data, and effectively visualising key metrics that connect to the research objectives. We also cover best practices for integrating eye-tracking into impactful field studies across brick-and-mortar and online retail channels.

Step 1: determine if eye-tracking makes sense

The first and most vital step when designing a field study is to evaluate whether incorporating eye-tracking technology can provide additional insights necessary to address the core research questions. Tracking visual attention and orientation offers value, but requires thoughtful implementation to produce substantive new knowledge. In our assessment, eye-tracking in retail settings is most called for under conditions where other methods face limitations. Some of these situations include the following:

1. There is a need to capture and analyse complete shopper journeys and key decision processes within them, such as approximations of attention and active consideration. This proves challenging through shadowing or manual observations alone, whilst eye-tracking can easily capture some critical nuances to approximate these mechanisms. This is relevant for researchers who wish to use either qualitative or quantitative methods when using eye-tracking data.

	SMALLER-SCALE SAMPLES	LARGE-SCALE SAMPLES
INCIDENTAL EXPOSURE	 Ambulant Eye tracker	 Manual observations/CCTV (field)
FORCED EXPOSURE	 Stationary Eye tracker	 Online experiments Stationary Eye tracker/ Webcam Eye tracker

Fig. 2. Suggested eye-tracking choices.

- Pinpointing visual engagement with precise in-store details – such as price tags, promotional materials or seasonal displays – is necessary to answer the research question. Surveys, brief intercept interviews, shadowing, and other observational studies do not provide insights into what a shopper looks at. Questionnaires are particularly poor at capturing holistic visual processing as they require shoppers to remember conscious attention.
- The researched in-store components centre heavily on incidental exposure or could be influenced by ambient factors, making recall problematic and biased in self-reports. For example, suppose adding a new type of mannequin or end-cap display in a store is of interest. In that case, it may be essential to see if it genuinely captures more attention and consideration than other elements in the store that simultaneously compete for the shoppers' attention. Real-world tracking provides needed validity over lab simulations in these cases.

In essence, the suitability of eye-tracking data hinges on whether it will uncover important behavioural patterns and insights beyond what existing observational and survey-based techniques can suitably reveal. This is directly tied to the research objectives. If eye-tracking is not necessary for comprehensively addressing the questions, the significant resources and specialised expertise required will take significant efforts and resources without an appropriate return on investment. Aligning methodology to the nature of the research question and conceptual development of the research project remains critical from study conception to execution. Fig. 2 provides examples of suitable choices for eye-tracking based on whether the researcher is studying areas of retailing that rely on incidental or forced exposure. In the next step, where we discuss the study design phase, we will explain why the sample size also dictates what is practically feasible in an eye-tracking study.

Step 2: designing and planning the study

Because it is so resource-intensive, researchers should carefully consider the various aspects of running the study.

Here, we provide some critical steps in the planning and design phase that are useful for running an eye-tracking study.

Gaining access and building trust. Accessing natural retail settings requires substantial relationship-building and transparency about research objectives. Provide store managers with detailed information on the study goals, data collection and privacy protocols, potential risks or intrusions into everyday operations, and expected timeframe. Allow hands-on interaction with the ambulatory eye trackers so they understand what shoppers experience (e.g., [Minahan et al. 2013](#)). In larger organisations, it may be helpful to start with store manager approvals before seeking corporate permissions, so that one has specific advocates helping facilitate headquarters buy-in. Consider giving managers an overview of findings relevant to their location as a gesture of thanks for their cooperation.

Understanding eye-tracking technology. Ambulatory eye trackers use optical sensors that reflect infrared light off the cornea or forward-facing cameras tracking pupil movement. Units must be accurately calibrated for each user to ensure the validity of eye gaze data. Calibration involves having shoppers look at fixed points while the equipment aligns with their pupil location and movement patterns. Data collectors should practice calibration procedures extensively before fieldwork to ensure everything goes smoothly in the store. Additional training content should cover equipment assembly, participant instructions, positioning adjustments if glasses slip, troubleshooting connections or recording issues, and confirmation of accurate data capture. It is often a good idea to schedule office test sessions and run preliminary on-site tests. That is also an excellent opportunity to clarify to research assistants how and where the sampling should be done.

Defining store areas to be examined. Carefully outline which types of environments and store areas are essential for the research questions. Priority zones like merchandise displays or service counters can be mapped out for data collectors so that they only collect data in those areas. Targeted scoping thus allows more data collection sessions per day. If examining a sequence of spaces or the customer journey from the store entrance to checkout is crucial, then fuller trip tracking is warranted despite longer recordings. Balance depth/duration and participant numbers.

Making decisions on sample sizes. Larger sample sizes strengthen statistical analysis but are extremely difficult for resource-intensive ambulatory eye-tracking projects. Calculate the minimum sample needed for adequate power based on expected effect sizes and specific statistical tests. It is essential to factor in the relatively high variation among shoppers' eye movements. If large amounts of data are impractical to collect, exploring more qualitative research approaches may be an idea. Capturing supplemental data like sales figures or space usage metrics helps confirm eye-tracking findings if sampling issues arise. Again, because it takes a relatively long time to collect many eye-tracking videos in the field, it is crucial to have a realistic plan for how much data can be collected during a specific period. This is especially important to consider if one utilises a quasi-experimental design, such as having different in-store marketing during different weeks.

Data triangulation. Combining ambulatory eye-tracking with shopper receipts, brief interviews, or camera-based tracking can provide more context. For full shopping trips, an entrance and exit survey is often helpful in collecting additional insights. It also reminds shoppers to return the eye-tracking equipment. (Because the later versions are relatively unobtrusive, it is widespread that shoppers walk out of the store while forgetting that they are wearing them!) For more focused studies, such as those on a specific marketing stimulus, manual shopping observations are often helpful to get an alternative measure of the same key effect.

Ethics approval. Obtaining ethics/IRB approval is essential for in-store eye-tracking research because it collects biometric data from participants and potentially identifiable information about other people in the store environment. Ethics boards will likely want accountability procedures integrated into the research plans, demonstrating respect for shoppers' and employees' privacy throughout the project lifecycle. Building trust around sensitive data handling is vital. Researchers should have clear protocols addressing the following:

1. Informed consent procedures explaining how eye-gaze data will be utilised.
2. Steps taken during data collection and analysis to protect participant anonymity.
3. Secure storage of video files after analysis had concluded.
4. Protocol for obscuring identifying details for visualisations or video segments used in academic publications or presentations.
5. Confirmation that aggregated eye-tracking metrics will be the only data shared back with partner retailers, with no raw videos or individual heatmaps provided that could reveal identities.

Step 3: out in the field and collecting data

When the study is about to start, several considerations must be considered once it has commenced. We always recommend that the research team spend time with the data collection team at the beginning of each study to work out any quirks. At a bare minimum, the research team would do well

to spend at least the entire first day with them to ensure that they act in the intended way. This is partly due to the issues relating to collecting field eye-tracking data and partly the issues with retailing field studies and experiments.

Controlling the research design. Any manipulations to the store environment must align precisely with the intended research design on data collection days. For example, with a weekly quasi-experimental shelf layout change, the research assistant should confirm that the setup is implemented correctly each week. Instruct assistants to validate configurations multiple times, capturing any accidental changes by staff unaware of the study layout. Consider drafting detailed protocols for research assistants to maintain continuity across conditions systematically.

Preparing the setup. The research assistant should fully charge all ambulatory eye tracker batteries the night prior and label sets for each participant. Charged-out batteries can cost half a day of data collection! Have a set of backup batteries on hand, along with multiple spare SD cards. Conduct test recordings to ensure proper functioning and calibrate if needed. Designate an organised bag or storage kit for eye-tracking materials with a checklist for assistants to consult each morning. Prepare tip sheets for resolving standard device issues like loss of calibration, cabling failures or blurred footage based on equipment limitations discovered during piloting.

Sampling protocol. We advise against inviting participants to the store to participate specifically in the study, as this may take away the realism of a regular shopping trip and thus compromise the ecological validity field studies are designed to provide. Unless targeting specific shopper segments, random intercept sampling often works best. One way we like to use is to specify an entrance "drafting point" to observe foot traffic flow. Determine an appropriate sampling interval based on the estimated participation rate and total sample size goals accounting for attrition due to technical errors. The data collector may approach every n^{th} customer passing through that point about participating. It is good to cover different days and representative times in proportion to overall store traffic distributions, which can quickly be done based on historical data and consultations with the store manager. Instruct assistants to track refusal reasons and exclusion criteria and contact the lead researcher as early as possible in the study if there are essential refusal patterns to participation.

Participant briefing. The recruiting assistant should follow a standard script to effectively explain the study purpose, data uses, privacy protections and the option to withdraw anytime without penalty. Notify participants regarding the collection of first-person visuals from the tracker's perspective. In some ambulatory eye-tracking systems, audio is also collected, which should be clear to any participant. Provide takeaway cards with project contacts for later questions and specify any further data usage in academic publications or business recommendations. We experience that quite a few participants are genuinely interested in the findings.

Entrance interviews/surveys. Once a participant has decided to participate, give them an entry survey, in case there

is a wish to measure specific control mechanisms. This could, for example, be how much money they had planned to spend in the store or which products they had planned to buy (Grewal et al. 2020; 2023b). Ensure these questionnaires are numbered by the participant number in the recording or any other measure to match the survey to the correct recording easily. Often, we number entrance (and exit) surveys and give participants a big note with the number. That way, the number on the note will be recorded in the video, and one can take a photo of the final shopping receipt next to this numbered note to later match the different data sources in an easy way. Speaking of receipts, if one is interested in what shoppers buy, tell them twice to keep their receipts. It can be a good idea to write this reminder on the individually numbered notes that the researcher sends with them. These small tricks make combining survey and receipt data easier with eye-tracking measures.

During the recording. When participants are in the store, depending on the eye-tracking type, they can track their movements until the connection to the participants' device is too far away from the data collector. In situations where the recording is expected to take some time, such as when an entire shopping trip is part of the study, it is always wise to use the waiting time for other activities. This could include coding up paper-and-pen surveys and potential receipts that may have been collected. It is essential, though, to keep an eye on the participants as it is not uncommon that they forget that they are wearing the glasses. More than once, we have had to run after shoppers out to the parking lot to get the device back (which is also why we are ensuring the device is insured for field work by your institution/organisation!)

After the recording is finished. After the participant has finished the eye-tracking part of the study (i.e., been exposed to the study's targeted retail environment), the data collector should meet with each participant to conclude data collection and conduct exit activities. The goal during this period involves capturing relevant perceptions, emotions, attitudes, and explanatory attributes not discernible from eye-tracking footage alone. Administer questionnaires covering these self-reported metrics—often around purchase satisfaction/influences, basket size or overall experiences. While questionnaires are filled in, it is an ideal moment to photograph final sales receipts to determine the exact spend and items selected—especially if eye recording stops before checkout.

The exit questionnaire can measure essential boundary conditions and demographic factors, including personality items, informative shopping style indicators, or customer category traits useful for segmenting analytical findings. Collect sufficient details on the individual to inform and enrich eye-gaze visual patterns captured within the store context. Link responses to behaviours through matched participant identifier codes.

Once exiting, participants can also be debriefed on more details on the study purpose if partial disclosures regarding aims or manipulations were necessary earlier to prevent demand bias during the in-store session. This way, a multi-

faceted capture of the retail shopping journey emerges—from entrance intentions to in-store behaviours using eye-tracking technology to post-experience reactions and outcomes. Triangulating these data streams provides a more nuanced understanding of the consumer response cascade within the testing environment.

Step 4: coding data

Once all the data is collected, one time-consuming element of eye-tracking research is coding the data. With stationary eye trackers connected to screens, the software can automatically code areas of interest to quantify fixations, fixation lengths, and more. However, such auto-coding abilities are limited with ambulatory eye-tracking devices. While eye-tracking software packages have progressed in auto-coding specific scenes (e.g., endcap displays and mannequins), manual coding remains necessary for most variables in full shopping trips. We are, however, optimistic that such coding algorithms will vastly improve with new developments in the future.

By this stage, researchers should know what variables to code before starting. The list should be comprehensive, as adding codes later requires restarting. Critical variables in retail field studies often include gaze behaviours - the number of fixations on elements and the duration of looking at elements. Fixation counting only requires one code, no matter the length, while “times of interest” between two fixations require marking the start and stop times to calculate duration. Some codable elements are:

1. Specific, unique items in the store, such as a piece of in-store advertisements or a specific stock-keeping unit (SKU).
2. Grouped stimuli in the store, such as price tags, products, other people, and staff members.

The built-in auto-coder works relatively well to identify fixations for unique in-store items like ads or SKUs, but verifying accuracy is needed. While auto-coding all videos is time-consuming upfront, it can provide a good starting point for analysing certain elements. Again, if the research question is related to a particular element in the store, this will be the go-to way of analysis in the first stage.

For grouped stimuli like products, manual coding frame-by-frame is required to capture gazes accurately. This involves the coder looking through the videos and marking these pre-determined elements in purpose-built research software packages. This is often done by pressing one key (e.g. “G”) when the shopper starts looking at a product category and another key (e.g. “H”) when they stop. Timing between the key presses marks the gaze duration. With hundreds of long videos to review, this process takes weeks or months to complete manually. Unfortunately, the labour investment seems necessary currently to capture such fine-grained behavioural data. Advancements in machine learning may one



Fig. 3. Example of a heatmap and a scan path in a retail shelf.

day allow accurate automated coding of these complex real-world stimuli.

Analysing precise eye movements (saccades) between visual targets and the series of such movements (resulting in “scan paths”) may also be insightful but requires extensive effort compared to controlled eye-tracking lab setups. Though we have yet to identify any in-store field studies that examine search patterns through saccades, assessing the order and timing of visual transitions across products or choice options may have research value. This seems most feasible for focused product assessment rather than lengthy full shopping trips.

Finally, there are opportunities to code the data for qualitative approaches, such as ethnographic research. In this case, the choice of codes may differ substantially from mere fixations and saccade movements to allow researchers to derive the themes relevant to their specific research question.

Step 5: analyse and visualise findings

The final stage relates to how researchers analyse and visualise their findings. They first need to decide whether to use qualitative or quantitative methods for analysing the eye-tracking videos, as specified in their research design and coding scheme. After coding, each participant will have one row in the dataset with all coded variables. This can be matched with triangulated additional data collected beyond eye-tracking, such as surveys and receipts.

Traditional analysis approaches are applicable after coding is complete. The choice of analysis methods flows from the research design and coded variables. The coded variables can be analysed for experimental studies using statistical tests like *t*-tests and regressions. As with any non-causal consumer study, endogeneity concerns should be addressed for non-experimental designs.

Popular ways to visualise eye-tracking findings include attention maps like heat maps (Blascheck et al. 2017).

Heatmaps overlay colours onto stimuli to represent the attention different areas receive. For example, in retail settings, specific store photos can have heat maps showing where shoppers looked the most. This works well for lab studies with a small number of stimuli and is increasingly possible with mobile tracking. However, heatmaps only cover certain parts of the store environment, so they must be carefully selected to convey relevant information to a reader of the research project. An example is in the left illustration in Fig. 3, where red visualises the most attention.

Another approach is scan path visualisations, which show numbered sequences of fixations and saccades. An example is in the right-hand illustration in Fig. 3. These numbered dots illustrate the order in which shoppers look at different shelf elements and how long they dwelled at any specific fixation point (visualised by larger circles in Fig. 3). For instance, scan paths can reveal which items first attracted attention or between which products shoppers compared. The limitation is that scan paths are individual sequences, so researchers must select representative examples to illustrate findings.

Beyond attention maps, findings can also be communicated through graphs of key metrics like total fixation duration, time to first fixation, and number of fixations. The visualisations selected should match the specific research questions and analytical approach. A problem with these is that they are, by definition, individual, so it is up to the researcher to use their judgment to select representative alternatives. Multi-method visualisation allows for conveying both aggregated patterns and individual sequences for a comprehensive perspective.

Conclusion

This paper has explored the potential of eye-tracking to uncover new insights into in-store shopping behaviour. As we have seen, shoppers rely heavily on visual cues when shopping. Eye-tracking lets researchers see what attracts shoppers' attention and influences their choices. With technology be-

coming more portable and affordable, we expect eye-tracking to become an increasingly vital tool for understanding shopper behaviour. This paper highlights where eye-tracking shows the most promise as we advance and where challenges remain. We also identify opportunities for creative, rigorous uses of eye-tracking that can provide actionable insights for retailers in their never-ending quest to improve the shopping experience. Our goal is to inspire more exploration of how this novel method can enhance our understanding of in-store decision journeys.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude, an AI assistant trained by Anthropic PBC, and Grammarly, a language editing software, for language suggestions and copy-editing assistance. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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